FedMedSeg: Privacy-Preserving Federated Learning for Medical Image Segmentation

Abstract

This paper proposes FedMedSeg, a federated learning framework for privacy-preserving medical image segmentation. We evaluate on brain tumor segmentation using the BraTS 2023 dataset distributed across 5 simulated hospital nodes. FedMedSeg achieves a Dice score of 0.891, only 1.2% below the centralized training baseline of 0.903. Our framework uses a modified U-Net architecture with attention gates and implements Federated Averaging (FedAvg) with differential privacy guarantees (epsilon = 3.0). The implementation uses PyTorch 2.1 and the Flower federated learning framework version 1.5. Training was conducted on heterogeneous GPU clusters simulating real hospital environments. The key contribution is our Adaptive Aggregation Strategy (AAS) that handles non-IID data distributions across hospital sites.

1. Introduction

Medical image segmentation is crucial for clinical diagnosis, treatment planning, and surgical navigation. Deep learning has achieved remarkable results, but training requires large, centralized datasets · which conflicts with patient privacy regulations like HIPAA and GDPR. Federated learning offers a solution by enabling model training across distributed hospital sites without sharing raw patient data. The main objective of this paper is to develop a federated learning framework that achieves near-centralized accuracy for medical image segmentation while providing formal privacy guarantees through differential privacy. We focus on brain tumor segmentation as our primary task due to its clinical importance and the availability of standardized benchmarks. Our approach also uses transformer-based attention mechanisms within the U-Net architecture to improve segmentation of small tumor regions.

2. Related Work

Federated learning was introduced by McMahan et al. (2017) with the FedAvg algorithm. In medical imaging, Li et al. (2020) applied FedAvg to brain tumor segmentation achieving 0.85 Dice score. Sheller et al. (2020) demonstrated federated learning across 10 institutions for glioma segmentation. Privacy-preserving techniques include differential privacy (Abadi et al., 2016) and secure aggregation (Bonawitz et al., 2017). Recent works combine transformer-based architectures with federated learning · FedViT (Park et al., 2023) used vision transformers in federated settings for classification tasks. Our work extends this line of research by incorporating attention-gated U-Net within a differentially private federated framework, specifically targeting segmentation.

3. Methodology

FedMedSeg consists of four components. Component 1: Architecture - We use a modified U-Net with an encoder-decoder structure containing 4 downsampling and 4 upsampling blocks. Attention gates are inserted at each skip connection to focus on relevant regions. The encoder uses ResNet-34 blocks pretrained on ImageNet. Total parameters: 31.4 million. Component 2: Federated Training Protocol - Each hospital trains locally for 5 epochs per communication round using Adam optimizer with learning rate 1e-4. After local training, model updates (not data) are sent to the central server. We run 200 communication rounds total. Component 3: Adaptive Aggregation Strategy (AAS) - Instead of simple averaging, AAS weights each hospital contribution by the inverse of its local validation loss. Hospitals with better-performing local models contribute more to the global model. Mathematically: w_global = sum(alpha_i * w_i) where alpha_i = (1/loss_i) / sum(1/loss_j). Component 4: Differential Privacy - We add calibrated Gaussian noise to model updates before aggregation. The noise scale is set to achieve (epsilon=3.0, delta=1e-5)-differential privacy using the moments accountant method. Gradient clipping with max norm of 1.0 is applied before noise addition.

4. Dataset

We used the BraTS 2023 (Brain Tumor Segmentation) Challenge dataset. It contains 2,000 multi-modal MRI scans (T1, T1-Gd, T2, FLAIR) with expert annotations for three tumor sub-regions: enhancing tumor (ET), tumor core (TC), and whole tumor (WT). Each scan is a 3D volume of 240x240x155 voxels. We simulated 5 hospital sites by splitting the data using a Dirichlet distribution (alpha=0.5) to create realistic non-IID partitions. Hospital data sizes ranged from 280 to 520 scans. We used 80/10/10 train/validation/test split at each site. Data augmentation included random flipping, rotation (up to 15 degrees), and elastic deformation. The total dataset size was approximately 45GB across all modalities.

5. Results

FedMedSeg achieved Dice scores of 0.891 (whole tumor), 0.847 (tumor core), and 0.812 (enhancing tumor) on the combined test set. The centralized baseline achieved 0.903, 0.862, and 0.831 respectively. The gap between federated and centralized training was only 1.2-1.9% across all sub-regions. Without our AAS, standard FedAvg achieved 0.863 Dice (-2.8% vs centralized), demonstrating the value of adaptive aggregation. The differential privacy mechanism reduced accuracy by approximately 0.8% compared to federated training without privacy (0.899 Dice). Communication overhead was 2.1GB per round for the full model. Using gradient compression (top-10% sparsification), we reduced this to 210MB per round with only 0.3% accuracy loss. Training time was 72 hours across the 5-node cluster. The per-hospital training cost was approximately $500, totaling $2,500 for the entire federation plus $300 for the central server.

6. Privacy Analysis

We formally analyze the privacy guarantees of FedMedSeg. Using the moments accountant (Abadi et al., 2016), we achieve (epsilon=3.0, delta=1e-5)-differential privacy after 200 communication rounds with 5 local epochs each. The noise multiplier was set to 0.8 with gradient clipping at norm 1.0. We also evaluated membership inference attacks: the attacker achieved only 52.3% accuracy (near random chance of 50%), compared to 68.7% against the centralized model without privacy protection. This demonstrates strong practical privacy even against sophisticated attacks.

7. Conclusion

FedMedSeg demonstrates that privacy-preserving federated learning can achieve near-centralized accuracy for medical image segmentation. The adaptive aggregation strategy effectively handles non-IID hospital data, and differential privacy provides formal guarantees with minimal accuracy loss. Future work includes extending to other medical imaging tasks (retinal image analysis, chest X-ray classification), exploring personalized federated learning where each hospital maintains a customized model, and reducing communication costs through knowledge distillation.